CPE 231 – Algorithms (1/2026)

Homework I

1. (5 points) Design an algorithm that takes a list of n integers and produces an output list that cyclically right shift the input list by s positions, where $1 \leq s \leq n-1$. For example, suppose your input is $3, 4, 12, 9, 8, 7$ and $s = 3$, the output must be $9, 8, 7, 3, 4, 12$. Justify if it is a valid algorithm. You must not use an extra array other than that needed to store the input list.

2. (15 points) Write down three different algorithms to determine the greatest common divisor (gcd) of two numbers m and n. Then, discuss their pros and cons.

Algorithm gcd algorithm

1: Input: Two non-negative, not both zero, integers m and n ($m \geq n$)
2: Output: A largest integer not greater than $\min(m, n)$ that divides m and n

3. (10 points) From the algorithm below, work out the counts of basic operations $C(n)$ and its growth order.

Algorithm Mystery

1: **Input:** Positive integer n
2: **Output:**
3: $r \leftarrow 0$
4: **for** $i := 1$ to $n - 1$ **do**
5: **for** $j := i + 1$ to n **do**
6: **for** $k := 1$ to j **do**
7: $r \leftarrow r + 1$
8: **return** r

4. (15 points) From the given algorithm,

Algorithm YouTryFigureItOut

```
1: Input: An array of positive integers A[1...n]
2: Output:
3:
4: for i = 1 to n − 1 do
5:     for j = 1 to n − i do
6:         if A[j + 1] < A[j] then
7:             swap A[j] and A[j + 1]
8: i ← 1
9: highest ← 0
10: while i ≤ n do
11:     l ← 1, v ← A[i]
12:     while i + l ≤ n and A[i + l] = v do
13:         l ← l + 1
14:     if l > highest then
15:         highest ← l, m_v ← v
16:     i ← i + l
17: return m_v
```

(a) (3 points) What task does it solve ? Explain from the algorithm.

(b) (2 points) What are the basic operations in the algorithm ?

(c) (3 points) Are best-case and worst-case $C(n)$ (based on the key comparison) the same? When do they happen?

(d) (7 points) Determine $C(n)$ in each scenario in (c) and draw conclusion on the algorithm efficiency. Show the computation details.

For the nested while loop, $C_2(n) = n \in \Theta(n)$. Therefore,

$$C(n) = C_1(n) + C_2(n) \in \Theta(\max\{n^2, n\}) = \Theta(n^2)$$